

Space Age AI Solutions

Session Init Reference

Version 2 · Hook file: `.claude/hooks/session-start.sh` · Repo: `Burst37/Space-Age-Skills` · Built 2026-06-21

What This Document Is

This is the permanent offline reference for the Space Age AI Solutions Claude Code session initialization system. Keep it so you can rebuild the hook from scratch if Claude ever loses context or the repo is corrupted.

Every Claude Code session (web, desktop, GitHub Action) fires a **SessionStart hook** automatically. That hook does two things: (1) immediately injects a structured `additionalContext` string that tells Claude exactly what to do before answering any task, and (2) syncs all skills from the repo into `~/.claude/skills/` in the background so they're available as slash commands.

The hook was rebuilt from scratch during the session of 2026-06-21 after the v1 hook was producing a generic/stripped context. The v2 upgrade added the full 4-step hard gate, the tier 2 routing matrix, auto-trigger rules, and a more complete infra quick reference. It also fixed background task safety (guards against empty `SKILLS_DST`, conditional npm install, safe `CLAUDE_ENV_FILE` check).

How the Hook Works (Mechanics)

Hook Location & Format

The hook lives at `.claude/hooks/session-start.sh` in the repo root and is registered in `.claude/settings.json` under the `hooks.SessionStart` array. Claude Code executes it on session open.

Two-Phase Execution

Phase	What happens	Why
Phase 1 (blocking)	Prints a single JSON object to stdout containing <code>hookSpecificOutput.additionalContext</code> and <code>hookSpecificOutput.reloadSkills: true</code>	Claude reads this before the conversation starts — it becomes the system-level context that governs the whole session
Phase 2 (background &)	Syncs <code>.claude/skills/*</code> → <code>~/.claude/skills/</code> , appends env vars to <code>CLAUDE_ENV_FILE</code> , runs <code>npm install</code> if <code>package.json</code> exists	Non-blocking — session starts instantly, skills and env vars are ready within seconds

JSON Output Format

```
{
  "hookSpecificOutput": {
    "hookEventName": "SessionStart",
    "additionalContext": "... the full 4-step context string ...",
    "reloadSkills": true
  }
}
```

`reloadSkills: true` tells Claude Code to re-scan `~/.claude/skills/` and make all SKILL.md files available as slash commands immediately.

The 4-Step Hard Gate (injected into every session)

This is the exact content injected as `additionalContext`. Claude is blocked from responding to any user task until all 4 steps are confirmed complete.

Step 1 — Read Drive Memory

TOOL CALL SEQUENCE

```
mcp__Google_Drive__search_files:
  query: title = 'YYYY-MM-DD.md' and '1uimIv6Uou7Ug0bYabz_P4Ylr2LhVLIiU' in
  parents

→ read file with mcp__Google_Drive__read_file_content
→ if not found: get most recent file in folder, read that
→ load all context, pending tasks, key IDs from it
```

The **SESSION_MEMORY folder** (1uimIv6Uou7Ug0bYabz_P4YLR2LhVLIiU) is the primary memory layer — Google Drive, not the Obsidian vault (which lives on the Windows machine only, not accessible from VPS or web sessions).

Step 2 — Tier 0 Skills (all active, every session, no exceptions)

Skill	Mode	What it does
caveman	ACTIVE immediately	~75% token reduction. Terse responses. Never self-reverts. Only off on "stop caveman" or "normal mode"
superpowers	Spec-first	No code written without an approved spec. Enforces subagent-driven-development for multi-task plans
SpaceAge_Orchestrator_v2	Routing brain	5-tier capability registry. Consulted before any task. Silent execution — no manual invocation required
karpthy-guidelines	Code quality	Baseline code quality rules applied to all output
icm-workspace-architect	Structure	Workspace/directory structure decisions
sa-obsidian-vault-ops	Memory ops	Drive read at session start, Drive write at session end

Step 3 — Tier 2 Stack Routing

Signal in user request	Tier 2 stack loaded
bot / scraper / loyalty	LoyaltyBot — VPS 146.190.78.120 + pm2 + Node.js
website / Vercel / Next	Web — mcp__Vercel__* + Next.js + Tailwind
image / logo / design	Design — mcp__Adobe_for_creativity__* + mcp__Figma__*
data / pipeline / airtable	Data — mcp__Airtable__* + mcp__Coupler_io__*
VPS / server / deploy	Infra — SSH + pm2 + nginx
Obsidian / vault / notes	Obsidian — obsidian-skills + obsidian-cli + obsidian-markdown
doc / slides / report	Docs — quarkdown + mcp__Gamma__*
social / web search	Reach — agent-reach MCP
agent / parallel / plan	Agents — superpowers subagent-driven-development + cavecrew

Step 4 — Confirm Checklist

- ☐ Drive memory read (or confirmed empty)
- ☐ caveman ACTIVE full mode

- ☐ superpowers loaded
- ☐ SpaceAge_Orchestrator_v2 routing done
- ☐ Model routing: claude-opus-4-8 heavy / claude-haiku-4-5-20251001 light

❑ MCP > CLI > SDK > raw key priority confirmed

Auto-Trigger Rules (always on, no invocation needed)

Event	Auto-trigger
git commit	caveman-commit skill
PR review	caveman-review skill
bug / test failure	systematic-debugging BEFORE any fix
claiming done / PR ready	verification-before-completion FIRST
multi-task plan	subagent-driven-development
user pastes URL	defuddle (not WebFetch), unless URL ends in .md
spawning subagents	cavecrew for correct agent type

Infra Quick Reference

Key	Value
VPS	146.190.78.120
GitHub user	Burst37
Skills Repo	Burst37/Space-Age-Skills
Skills Drive folder	1XWYm8AhG83vMn1p3RpM1UAKmiKcsnoC9
Session Memory folder	1uimIv6Uou7Ug0bYabz_P4Ylr2LhVLIu
Vercel Team	team_b7Ju9bt8GNoiLnMor6ieC8J7
Agent Dashboard	http://146.190.78.120:3000
API Proxy	http://146.190.78.120:3400
Model (heavy tasks)	claude-opus-4-8
Model (fast/cheap)	claude-haiku-4-5-20251001
Obsidian vault location	Windows machine ONLY — NOT on VPS. Drive is the memory layer.

Anti-Forget Rules

Rule	Detail
CapSolver key	Check Drive LoyaltyBot note BEFORE asking user
Vercel	Use MCP or CLI. Never ask for token.
GitHub	Use mcp__github__* or gh CLI
Credentials	Log NAMES only in Drive notes. NEVER values.
caveman	Never self-revert. Only off on: "stop caveman" / "normal mode"
LoyaltyBot	Standalone product. ZERO connection to website pipeline. Do not conflate.
Encore logo	Upper left chest in ALL generated images
Skill naming	lowercase-hyphen only. NO SA- prefix.

Env Vars Written Each Session

These are appended to `CLAUDE_ENV_FILE` in the background phase so they're available in any shell during the session:

```
SA_SESSION_MEMORY_FOLDER="1uimIv6Uou7Ug0bYabz_P4Ylr2LhVLIu"
SA_SKILLS_DRIVE_FOLDER="1XWYm8AhG83vMn1p3RpM1UAKmiKcsnoC9"
SA_VPS_HOST="146.190.78.120"
SA_GITHUB_USER="Burst37"
SA_SKILLS_REPO="Burst37/Space-Age-Skills"
SA_VERCEL_TEAM="team_b7Ju9bt8GNoiLnMor6ieC8J7"
SA_MODEL_HEAVY="claude-opus-4-8"
SA_MODEL_LIGHT="claude-haiku-4-5-20251001"
```

How to Rebuild From Scratch

If the repo is lost, the hook is corrupted, or Claude starts a session without any context — use these steps to restore everything.

1. Restore the hook file

Create `.claude/hooks/session-start.sh` from the full source below and `chmod +x` it.

2. Register it in settings

In `.claude/settings.json`, ensure the hooks block includes:

```
{
  "hooks": {
    "SessionStart": [
      {
        "hooks": [
          {
            "type": "command",
            "command": "bash .claude/hooks/session-start.sh"
          }
        ]
      }
    ]
  }
}
```

3. Rebuild the skills

Each skill is a folder in `.claude/skills/<name>/SKILL.md`. The 36 skills currently installed are:

SpaceAge_Orchestrator_v2	three-brain	caveman
superpowers	karpthy-guidelines (TBD)	icm-workspace-architect
(TBD)		
sa-obsidian-vault-ops	spaceage-savo-creative-director-os	
video-intelligence	website-intelligence	design-taste-frontend
design-motion-principles	universal-build-handoff	sa-stop-slop
sa-anti-slop-writer	sa-voice-agent-builder	sa-swarm-orchestrator
sa-local-seo-geo	sa-graphify-operator	sa-pattern-genome
sa-hero-genome	sa-gsap-motion-library	sa-ecommerce-genome
sa-qa-production	carousel-skill	motion-graphics-skill
static-visual-skill	video-editor-skill	gpt-image-prompt-skill
quarkdown	defuddle	json-canvas
agent-reach	session-start-hook	obsidian-skills
obsidian-cli	obsidian-markdown	

4. Restore the memory layer

The **Google Drive folder** `1uimIv6Uou7Ug0bYabz_P4YLR2LhVLIiU` holds daily session notes in `YYYY-MM-DD.md` format. These are the long-term memory. The Drive folder ID never changes even if files are renamed.

5. Push and open a new session

Commit the restored hook + skills to branch `main` (or the active dev branch), push to `Burst37/Space-Age-Skills`, and start a new Claude Code session. The hook fires automatically and all 4 steps run before any task.

Full Hook Source (session-start.sh v2)

```
#!/bin/bash
# Space Age AI Solutions – Session Start Hook v2
# Emits additionalContext FIRST (non-blocking), then syncs skills in background.
set -euo pipefail
```

```
if [ "${CLAUDE_CODE_REMOTE:-}" ≠ "true" ]; then
    exit 0
fi
```

```
TODAY=$(date +%Y-%m-%d)
DOW=$(date +%A)
```

```
CONTEXT="
||  SPACE AGE SESSION INIT v2 – ${TODAY} (${DOW})
||
"
```

HARD GATE: Complete all 4 steps before responding to any user task.

— STEP 1 – READ DRIVE MEMORY —

```
mcp__Google_Drive__search_files:
    query: title = '${TODAY}.md' and '1vimIv6Uou7Ug0bYabz_P4Ylr2LhVLIiU' in
parents
    → read file with mcp__Google_Drive__read_file_content
    → if not found: get most recent file in folder, read that
    → load all context, pending tasks, key IDs from it
```

— STEP 2 – TIER 0 SKILLS (ALL active, every session) —

```
caveman          → ACTIVE NOW. Full mode. Every response. No revert.
superpowers       → Spec-first. No code without approved spec.
SpaceAge_Orchestrator_v2 → Routing brain. Consult before any task.
karpathy-guidelines → Code quality baseline on all output.
icm-workspace-architect → Workspace structure decisions.
sa-obsidian-vault-ops → Drive memory ops (read start, write end).
```

— STEP 3 – ROUTE TO TIER 2 STACK —

```
bot/scraper/loyalty    → LoyaltyBot (VPS 146.190.78.120 + pm2 + Node.js)
website/Vercel/Next    → Web (mcp__Vercel__* + Next.js + Tailwind)
image/logo/design     → Design (mcp__Adobe_for_creativity__* + mcp__Figma__*)
data/pipeline/base     → Data (mcp__Airtable__* + mcp__Coupler_io__*)
VPS/server/deploy      → Infra (SSH + pm2 + nginx)
Obsidian/vault/notes   → Obsidian stack (obsidian-skills + obsidian-cli +
obsidian-markdown)
doc/slides/report      → Docs (quarkdown + mcp__Gamma__*)
social/web search      → Reach (agent-reach MCP)
agent/parallel/plan    → Agents (superpowers subagent-driven-development +
cavecrew)
```

— STEP 4 – CONFIRM CHECKLIST —

```
[ ] Drive memory read (or confirmed empty)
[ ] caveman ACTIVE full mode
[ ] superpowers loaded
[ ] SpaceAge_Orchestrator_v2 routing done
[ ] Model routing: claude-opus-4-8 heavy / claude-haiku-4-5-20251001 light
[ ] MCP > CLI > SDK > raw key priority confirmed
```

— AUTO-TRIGGER RULES (always on, no invocation needed) —

git commit → caveman-commit skill
PR review → caveman-review skill
bug / test failure → systematic-debugging BEFORE any fix
claiming done / PR → verification-before-completion FIRST
multi-task plan → subagent-driven-development
user pastes URL → defuddle (not WebFetch), unless .md
spawning subagents → cavecrew for correct type

— INFRA QUICK REF —

VPS: 146.190.78.120
GitHub: Burst37
Skills Repo: Burst37/Space-Age-Skills
Skills Drive: 1XWYm8AhG83vMn1p3RpM1UAKmiKcsnoC9
SESSION_MEMORY: 1uimIv6Uou7Ug0bYabz_P4Ylr2LhVLIu
Vercel Team: team_b7Ju9bt8GNoiLnMor6ieC8J7
Agent Dashboard: <http://146.190.78.120:3000>
API Proxy: <http://146.190.78.120:3400>
Model heavy: claude-opus-4-8
Model light: claude-haiku-4-5-20251001
Obsidian vault: Windows machine ONLY — NOT on VPS. Drive is memory.

— ANTI-FORGET RULES —

CapSolver key → check Drive LoyaltyBot note BEFORE asking user
Vercel → MCP or CLI, never ask for token
GitHub → mcp_github_* or gh CLI
Credentials → log NAMES only in Drive. NEVER values.
caveman → never self-revert. Off only on: 'stop caveman' / 'normal mode'
LoyaltyBot → standalone. ZERO connection to website pipeline.
Encore logo → upper left chest in ALL generated images.
Skill naming → lowercase-hyphen only. NO SA- prefix.

ONLY AFTER ALL 4 STEPS: respond to user task."

— EMIT JSON — must be first and only stdout

```
printf '%s' "{\"hookSpecificOutput\": {\"hookEventName\": \"SessionStart\",  
\"additionalContext\": $(printf '%s' \"$CONTEXT\" | python3 -c 'import json,sys;  
print(json.dumps(sys.stdin.read()))'), \"reloadSkills\": true}}\""
```

— BACKGROUND TASKS —

```
(  
  SKILLS_SRC="$CLAUDE_PROJECT_DIR/.claude/skills"  
  SKILLS_DST="$HOME/.claude/skills"  
  
  if [ -d "$SKILLS_SRC" ]; then  
    mkdir -p "$SKILLS_DST"  
    for skill_dir in "$SKILLS_SRC"/*; do  
      [ -d "$skill_dir" ] || continue  
      skill_name=$(basename "$skill_dir")  
      rm -rf "${SKILLS_DST:?}/${skill_name}"  
      cp -r "$skill_dir" "$SKILLS_DST/$skill_name"  
    done  
  fi  
  
  if [ -n "${CLAUDE_ENV_FILE:-}" ]; then  
    cat >> "$CLAUDE_ENV_FILE" << 'ENVEOF'  
export SA_SESSION_MEMORY_FOLDER="1uimIv6Uou7Ug0bYabz_P4Ylr2LhVLIu"  
export SA_SKILLS_DRIVE_FOLDER="1XWYm8AhG83vMn1p3RpM1UAKmiKcsnoC9"
```

```
export SA_VPS_HOST="146.190.78.120"
export SA_GITHUB_USER="Burst37"
export SA_SKILLS_REPO="Burst37/Space-Age-Skills"
export SA_VERCEL_TEAM="team_b7Ju9bt8GNoiLnMor6ieC8J7"
export SA_MODEL_HEAVY="claude-opus-4-8"
export SA_MODEL_LIGHT="claude-haiku-4-5-20251001"
ENVEOF
fi

if [ -f "$CLAUDE_PROJECT_DIR/package.json" ]; then
  npm install --silent 2>/dev/null || true
fi
) &
```